

storage system, which is mainly a hard drive. The data storage capacity depends on the hard drive, and the amount of data storage needed depends on the applications running the virtual machines. Because the virtual machines operate directly on SoCs and storage is directly connected, the system can run high performance databases very rapidly. And the decentralisation makes this system far simpler.

The supported hardware recommended to run these services are AMD CPUs from ASRock Industrial Computer Corp. Two of the recommended hardware are the 4800U motherboard and the V2000M consisting of up to 16 threads. With these, getting started with SynchronKnot solutions is easy. This hardware has certain unique features like accessing BIOS remotely.

The hardware is an embedded motherboard. The reason for calling it an SoC is its 10.2cm x 10.2cm (4x4 inch) form factor. You can directly install your operating system on this SoC. You can use any AMD x64 CPU for installing SynchronKnot services, but the aforementioned hardware is certified by SynchronKnot and is highly recommended. Also, a thick robust power supply will be required to run SynchronKnot services on other embedded hardware.

In a nutshell, this is a complete cloud solution on a SoC. You need an external memory storage solution, but as it is directly connected, data transfer is very fast. Moreover, no disparate hardware like switches and routers are required to implement this architecture.

## Who are these services intended for?

Anyone can choose this solution for personal usage instead of buying cloud services like Amazon AWS and Google Cloud. The threads and cores in SoCs can be separated to provide access to tenants, who can be partners or customers. SynchronKnot can be stacked

anywhere in the world and can remotely access virtual machines as tenants.

Services can also be provided based on SynchronKnot infrastructure. For this, one has to become a SynchronKnot certified systems integrator. One can then provide SynchronKnot solutions and hardware to clients, and guide them through.

## The learning curve for adopting this new technology

Let's say, an organisation uses centralised management facilities such as vCenter, Microsoft Virtual Machine Manager or Xen locally, and runs virtual machines on them. The next layer is connection with cloud services. With SynchronKnot, things get much simpler. It takes 12 seconds to install on a 2-core machine. After installation, one can log in from the graphical user interface (GUI) itself, and all other virtual machines logged in as tenants can be seen on the GUI.

With this solution, one can mirror the virtual machines on another site. No extra skills or knowledge are required for this. The data manipulation that has to be done while using cloud services in the

database for proper GUI is not required here. Almost everything is automatic, and there is no need to go to the command lines unless spatial clusters are being done for the very first time.

The OS is Debian, which is one of the most secure operating systems. Once the OS is installed, one needs to follow the demo videos from <https://synchronknot.cloud/> to install the entire SynchronKnot within seconds. One can stack it up, and there is no need to configure anything in terms of switching, physically or virtually. One can connect ports in any topology such as star, tree, etc.

Therefore, an organisation can actually transition to SynchronKnot without having to change anything, because it supports VMware and all other standard OpenStack virtualisation technologies.

The whole setup is scalable — virtual machines can be added by adding hardware modules. For installation, Ethernet cable is required, and the GUI automatically loads up in added hardware. All that is needed is to select SynchronKnot from the GUI and everything gets aligned.

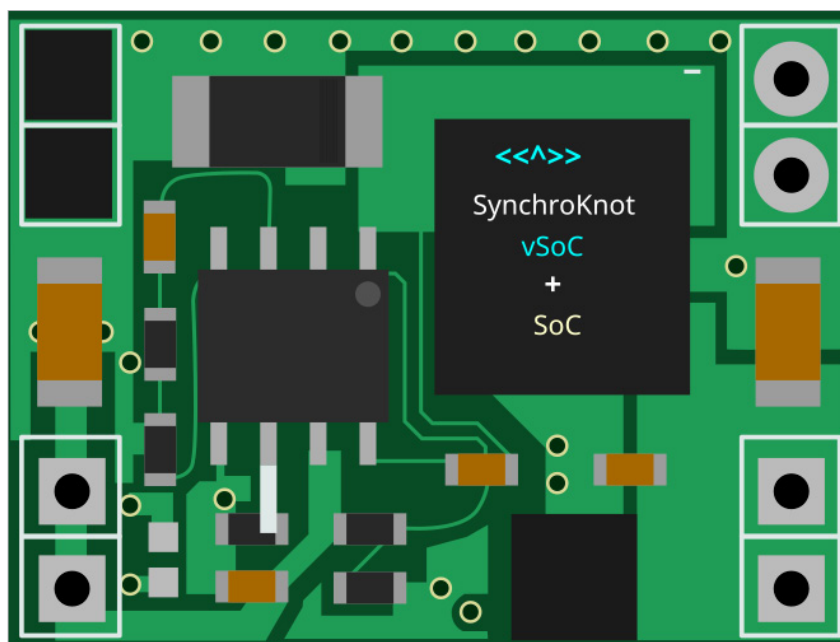


Figure 2: SynchronKnot vSoC (Image courtesy of SynchronKnot)